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Introduction

This interactive lesson walks you through the complete derivation of the **3-DOF Equations of Motion** for atmospheric reentry vehicles.

WHAT WE'LL BUILD

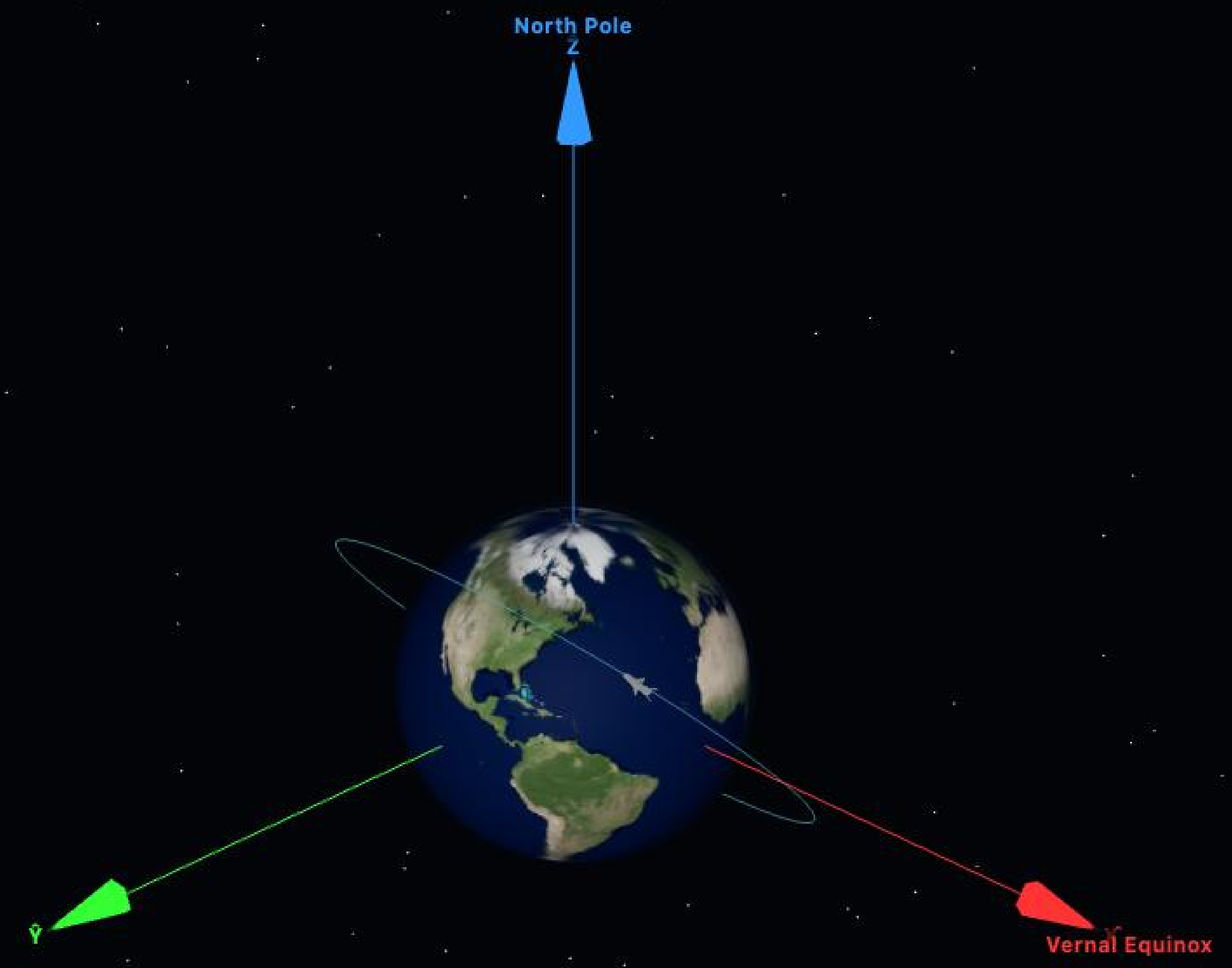
- Four nested coordinate frames
- Kinematic equations linking position angles to velocity
- All forces acting on the entry vehicle
- Six coupled scalar ODEs that govern the trajectory

CONTROLS

- ← → or buttons below — advance slides
- **Mouse drag** — orbit the 3D scene
- **Scroll** — zoom in/out

MECH 637: Astrodynamic Reentry · Fall 2022

Geocentric-Equatorial Frame (ECI)



The ECI Frame (**OXYZ**) is our inertial reference — fixed in space, origin at Earth's center.

AXIS DEFINITIONS

$$\hat{e}_x \rightarrow \text{Vernal Equinox}$$

$$\hat{e}_z \rightarrow \text{North Pole}$$

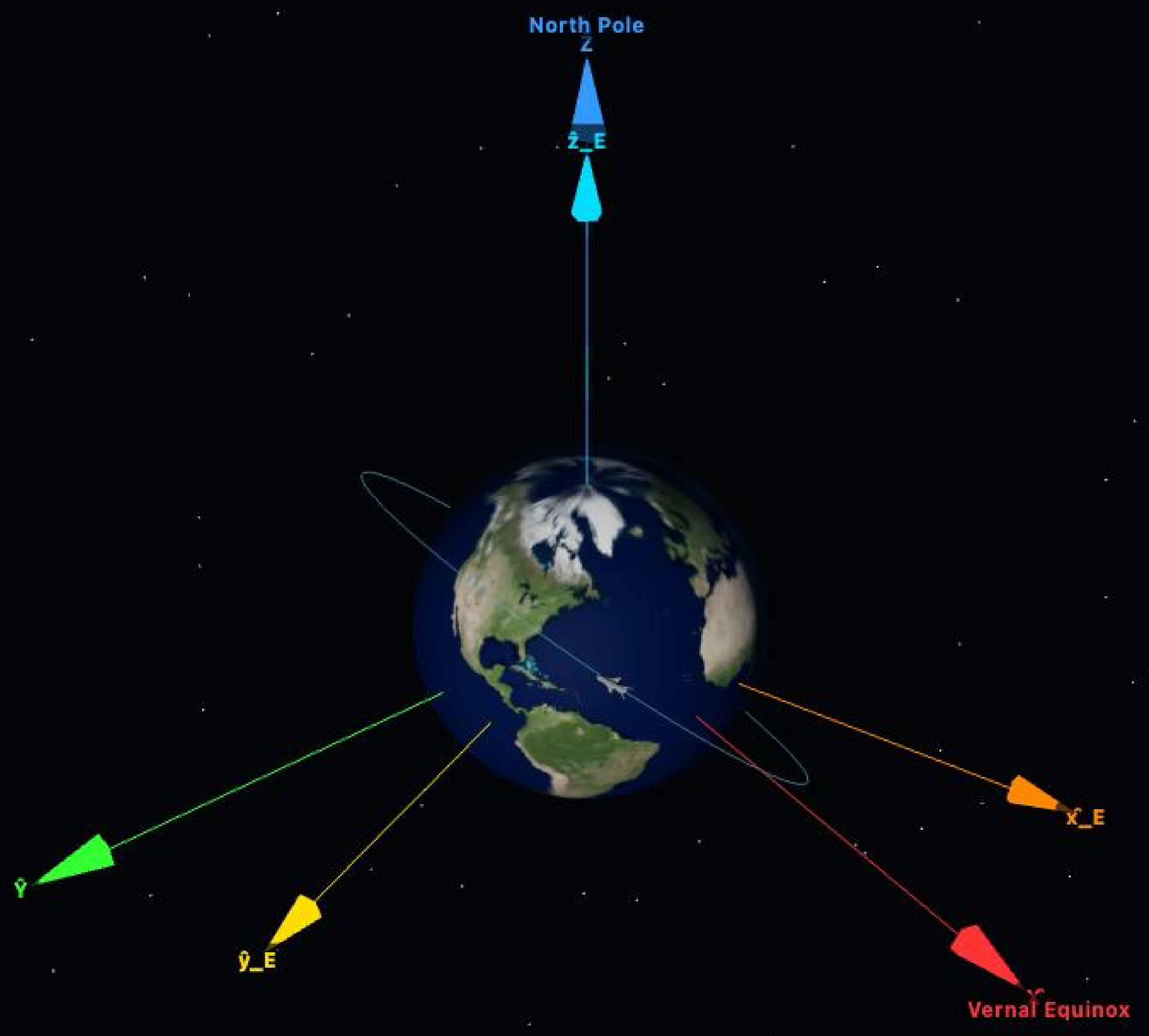
$$\hat{e}_y = \hat{e}_z \times \hat{e}_x \quad \text{(completes right-hand system)}$$

Because this frame does *not* rotate with Earth, Newton's Second Law holds in its standard inertial form:

$$m\ddot{\vec{r}}_I = \vec{F}_{\text{total}}$$

All subsequent frames — $OX_1Y_1Z_1$, $OX_2Y_2Z_2$, $OX''Y''Z''$ — are defined by rotations relative to OXYZ. The DCMs on slide 14 give the full transformation chain.

Planet-Fixed Frame ($OX_1Y_1Z_1$)



The **Planet-Fixed Frame** (**$OX_1Y_1Z_1$**,) rotates with Earth at constant angular velocity $\omega_\oplus$ about the polar axis.

EARTH'S ROTATION RATE

$$\omega_\oplus = 7.292 \times 10^{-5}\ \mathrm{rad/s}$$

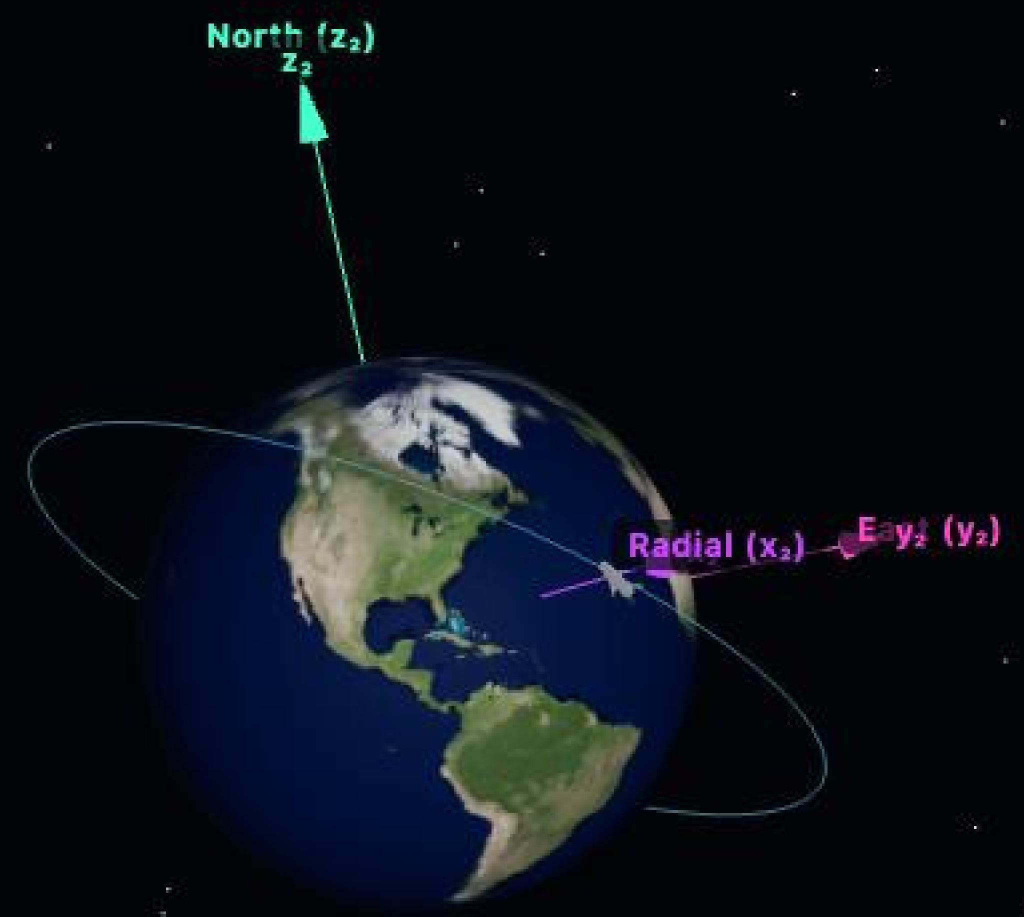
DCM — OXYZ → $OX_1Y_1Z_1$

$$[\hat{e}_1] = R_z(\omega_\oplus \Delta t)\,[\hat{e}]$$

$$R_z(\omega_\oplus \Delta t) = \begin{bmatrix} \cos(\omega_\oplus \Delta t) & \sin(\omega_\oplus \Delta t) & 0 \\ -\sin(\omega_\oplus \Delta t) & \cos(\omega_\oplus \Delta t) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The scene is frozen at a moment in time — the **amber $OX_1Y_1Z_1$ axes** are offset from the **white OXYZ axes** by the current Earth rotation angle $\omega_\oplus \Delta t$. As Earth spins, this offset grows continuously.

Vehicle-Pointing Frame ($OX_2Y_2Z_2$)



The **Vehicle-Pointing Frame** ($OX_2Y_2Z_2$) has its **origin at the planetary center**, with axes locked to the vehicle's instantaneous position.

AXIS DEFINITIONS

$$\hat{e}_{x_2} = \frac{\vec{r}}{|\vec{r}|} \quad \text{(radial — along position vector)}$$

$$\hat{e}_{y_2} = \hat{e}_z \times \hat{e}_{x_2} \quad \text{(east — parallel to equatorial plane)}$$

$$\hat{e}_{z_2} = \hat{e}_{x_2} \times \hat{e}_{y_2} \quad \text{(cross-track — completes RH system)}$$

ECEF → VEHICLE-POINTING ROTATION

$$[\hat{e}_2] = R_{y_2}(-\phi)\, R_{z_1}(\theta)\, [\hat{e}_1]$$

RST SHORTHAND USED IN THIS PRESENTATION

Later slides write this frame as **RST** — a compact alias:

$\hat{R}$	$\hat{S}$	$\hat{T}$
$= x_2$ = Radial	$= y_2$ = East	$= z_2$ = Cross-track

Any time you see a force vector with subscript *RST*, its three components are along $\hat{R}$, $\hat{S}$, $\hat{T}$ = x_2, y_2, z_2 of the Vehicle-Pointing Frame.

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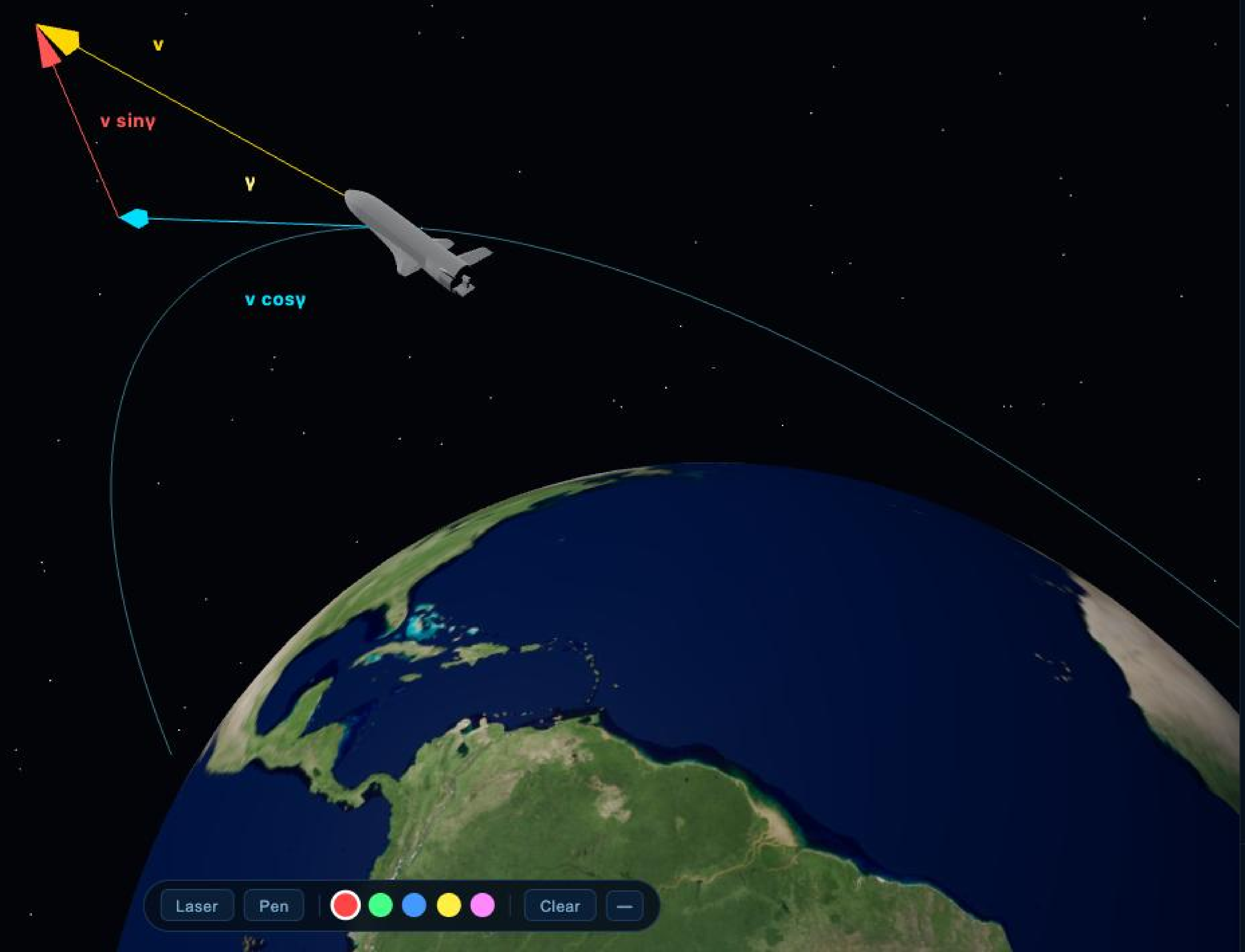
Geocentric Longitude λ & Latitude φ

The vehicle's position on the rotating planet is described by **geocentric longitude** λ and **geocentric latitude** ϕ.

Watch the **velocity vector** on this **eccentric orbit** ($e = 0.3$) — it swings direction *and* grows longer through periapsis, shrinks near apoapsis. This is Kepler's second law in action.

Press **Next →** to decompose *v* into its λ and ϕ rate components.



Velocity Decomposition — Radial & Horizontal

STEP 1 — SPLIT V BY FLIGHT-PATH ANGLE Γ

$$\underbrace{v\sin\gamma}_{\dot r\,=\,\text{radial rate}} \qquad \underbrace{v\cos\gamma}_{v_h\,=\,\text{horizontal speed}}$$

Velocity Decomposition — Heading Split

STEP 2 — SPLIT V COSΓ BY HEADING ANGLE Ψ

$$v_{\text{east}} = v\cos\gamma\cos\psi \qquad v_{\text{north}} = v\cos\gamma\sin\psi$$

STEP 3 — CONVERT SPEED TO ANGULAR RATE

$$\dot{\phi} = \frac{v_{\text{north}}}{r} = \frac{v\cos\gamma\sin\psi}{r}$$

$$\dot{\lambda} = \frac{v_{\text{east}}}{r\cos\phi} = \frac{v\cos\gamma\cos\psi}{r\cos\phi}$$

The extra $\cos\phi$ in $\dot{\lambda}$: a latitude circle at ϕ has radius $r\cos\phi$ — same eastward speed covers more λ degrees near the equator than near the poles.

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Flight-Path Angle γ

The **flight-path angle** γ is the angle between the *local horizontal plane* and the velocity vector.

RADIAL RATE

$$\dot{r} = v \sin \gamma$$

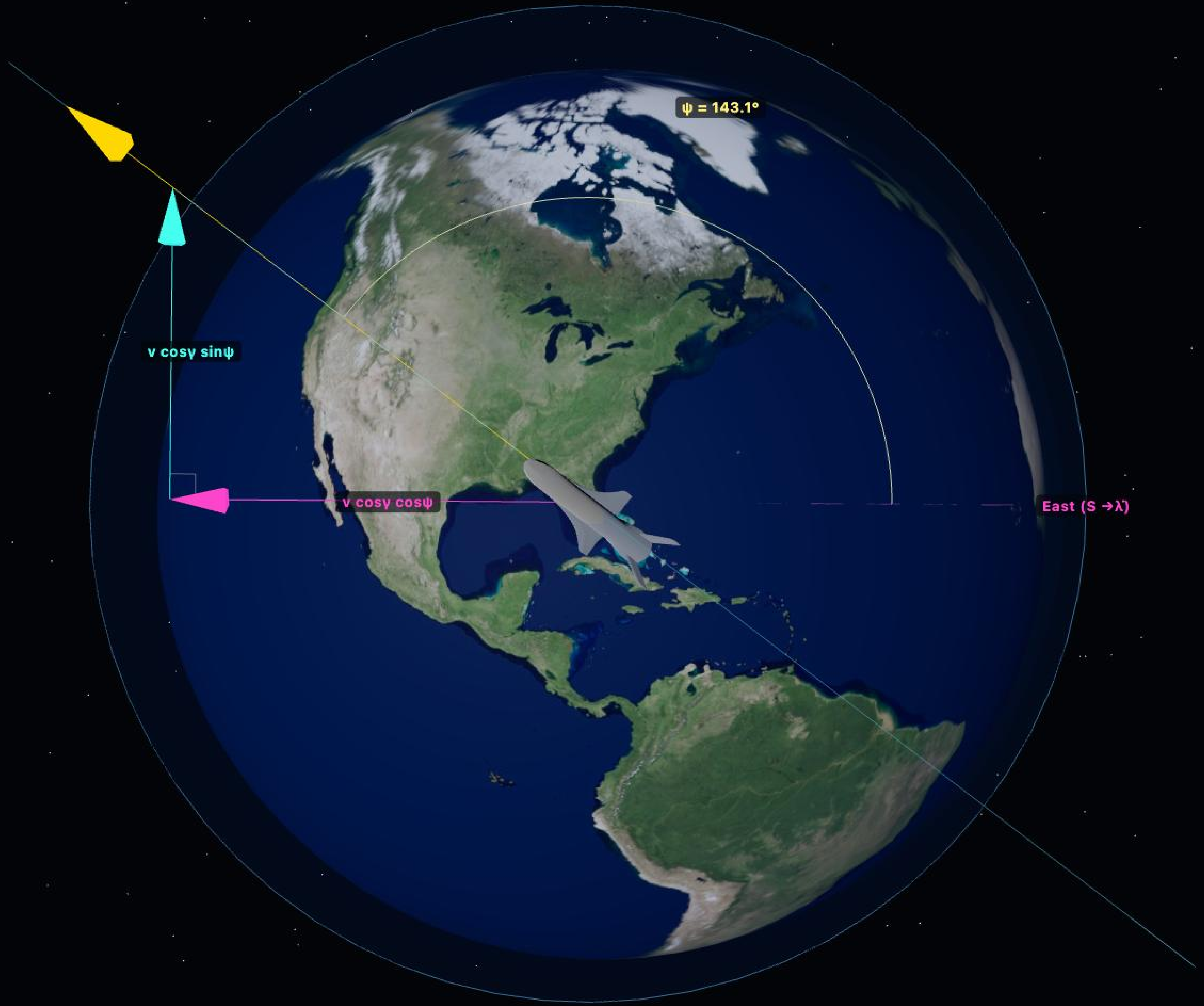
- $\gamma > 0$ — climbing (velocity above local horizontal)
- $\gamma < 0$ — descending (entry corridor)
- $\gamma = 0$ — purely horizontal flight

Watch the gold velocity arrow sweep as γ varies.

LIVE FLIGHT-PATH ANGLE

γ = +22.8°

↑ Climbing



Heading Angle ψ

The **heading angle** ψ describes the direction of horizontal flight — measured from East (S) toward North (T).

WHERE EACH ARROW COMES FROM

$v \cos\gamma$ = horizontal speed → split by ψ:

$$\dot{\lambda} = \frac{v\cos\gamma\,\cos\psi}{r\cos\phi} \quad \leftarrow \text{east component}$$

$$\dot{\phi} = \frac{v\cos\gamma\,\sin\psi}{r} \quad \leftarrow \text{north component}$$

- $\psi = 0^\circ$: all eastward → only $\dot{\lambda}$ grows
- $\psi = 90^\circ$: all northward → only $\dot{\phi}$ grows

The arrows in the scene are the two components — their lengths are proportional to cos ψ and sin ψ.

Velocity-Referenced Frame (VRF)

The **VRF** (**OX″Y″Z″**) is obtained from the Vehicle-Pointing Frame **$OX_2Y_2Z_2$** by two successive rotations defined by the velocity vector direction.

TWO-STEP ROTATION FROM $OX_2Y_2Z_2$

$$\text{Step 1: rotate } \psi \text{ about } \hat{e}_{x_2} = \hat{R} \quad \text{(heading azimuth)}$$

$$\text{Step 2: rotate } -\gamma \text{ about new } z' \quad \text{(flight-path angle)}$$

$$\mathbf{T}_{\mathrm{OX_2 \to OX''}} = R_{z'}(-\gamma)\, R_{x_2}(\psi)$$

AXIS DEFINITIONS (THIS PRESENTATION)

$$\hat{x}_v \parallel \vec{v} \quad \text{(along velocity)}$$

$$\hat{z}_v \perp \vec{v},\ \hat{z}_v \text{ lies in local horizontal plane}$$

$$\hat{y}_v = \hat{z}_v \times \hat{x}_v \quad \text{(completes right-hand system)}$$

Convention note: Velocity is along $\hat{x}_v$. All force equations use this convention consistently.

Forces expressed in VRF use frame-native scalars (D, L, T), then rotated back to RST via the transpose of this rotation chain.

TOGGLE FRAMES

• RST • VRF • Velocity

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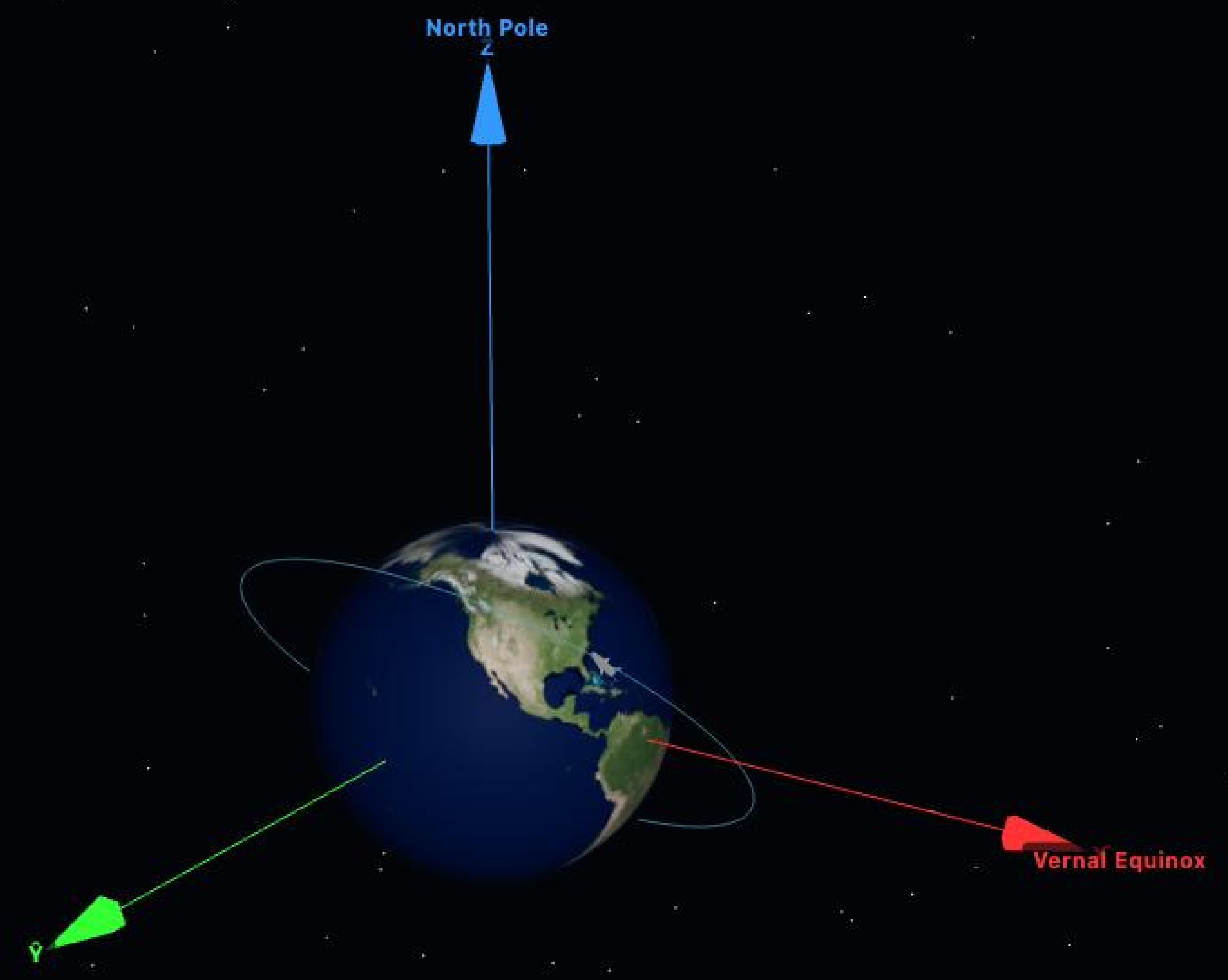

Coordinate Frame Transformations

Three successive Direction Cosine Matrices (DCMs) carry any vector from body to inertial space. Each 3×3 orthogonal matrix has inverse equal to its transpose.

$OXYZ$	$R_z(\omega_\oplus \Delta t)$ — Earth spin	$OX_1Y_1Z_1$
$OX_1Y_1Z_1$	$R_{y_2}(-\phi)\, R_{z_1}(\theta)$ — Vehicle-Pointing	$OX_2Y_2Z_2$
$OX_2Y_2Z_2$	$R_{z'}(-\gamma)\, R_{x_2}(\psi)$ — Velocity-Referenced	$OX''Y''Z''$

Press **Next** → to step through each animated DCM.

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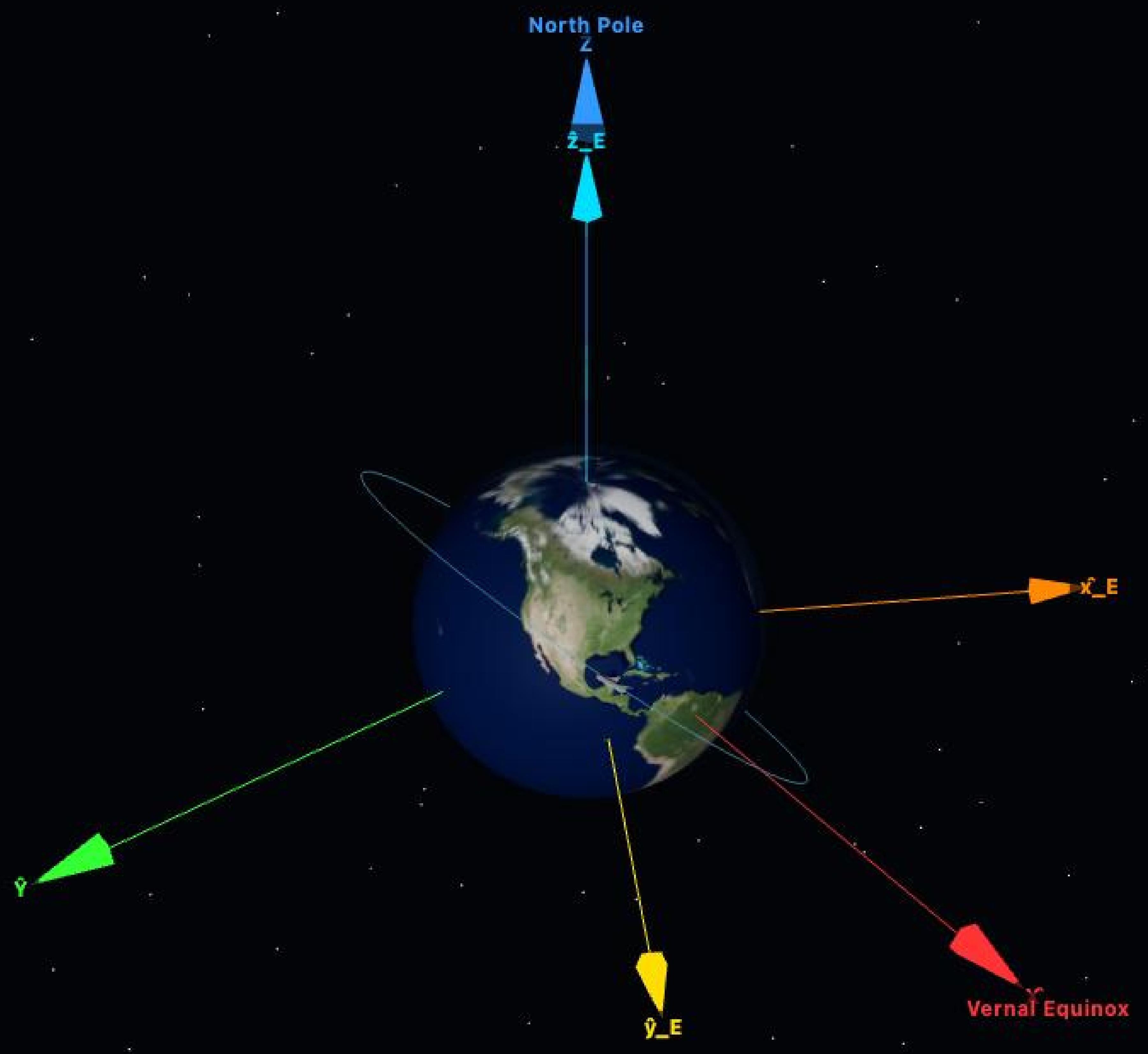
← Prev Next →

$OXYZ \rightarrow OX_1Y_1Z_1$: Earth Spin



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$OXYZ \rightarrow OX_1Y_1Z_1$ — EARTH SPIN

A single rotation about the polar $\hat{z}$-axis by $\omega_\oplus \Delta t$ carries $OX_1Y_1Z_1$ away from OXYZ. The amber axes spin with Earth.

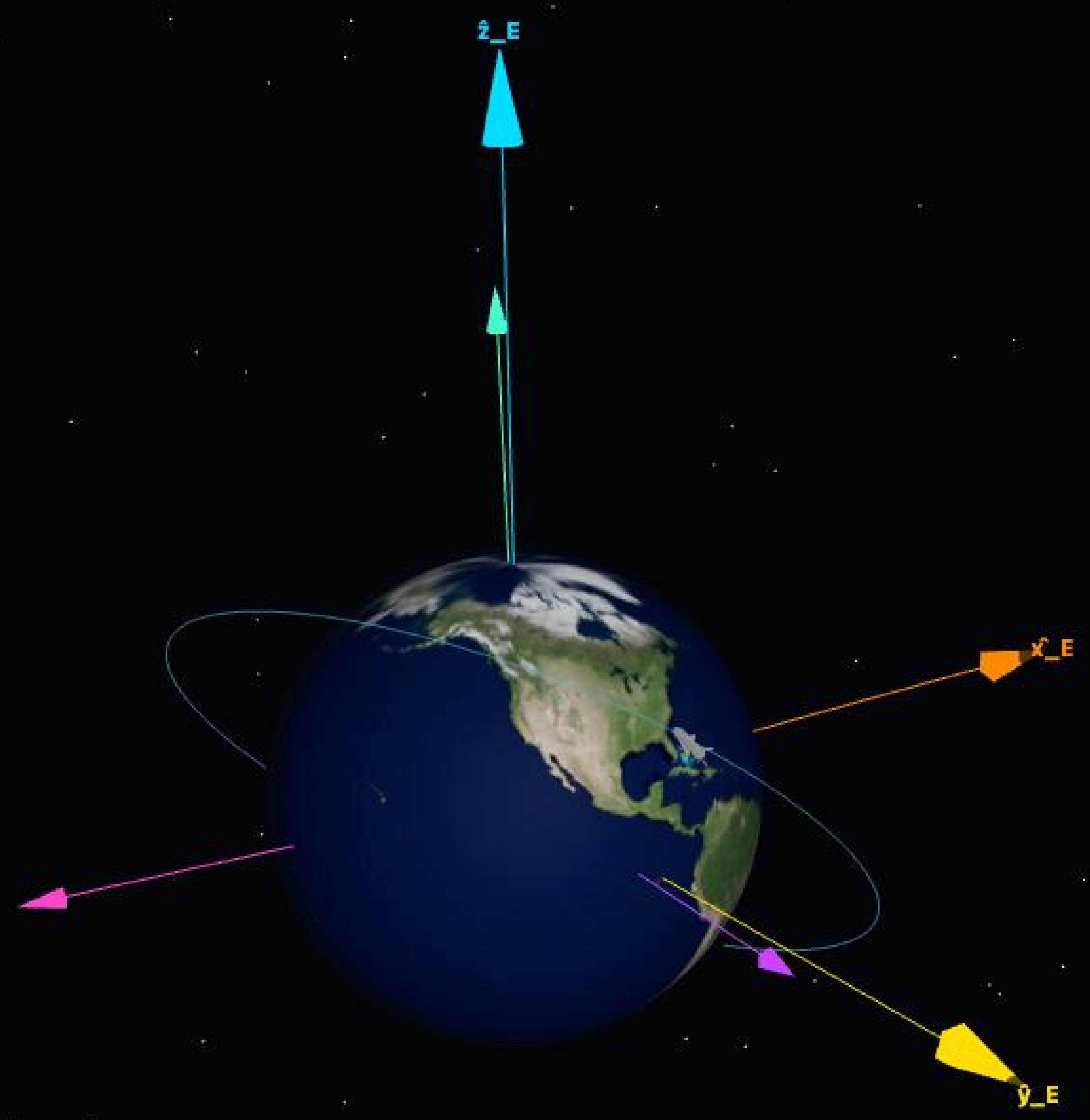
DIRECTION COSINE MATRIX

$$[\hat{e}_1] = R_z(\omega_\oplus \Delta t)\,[\hat{e}]$$

$$= \begin{bmatrix} \cos(\omega_\oplus \Delta t) & \sin(\omega_\oplus \Delta t) & 0 \\ -\sin(\omega_\oplus \Delta t) & \cos(\omega_\oplus \Delta t) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

LIVE ANGLE $\omega_\oplus \Delta t = 60.0°$

$OX_1Y_1Z_1 \rightarrow OX_2Y_2Z_2$: Vehicle-Pointing



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$OX_1Y_1Z_1 \rightarrow OX_2Y_2Z_2$ — VEHICLE-POINTING

Two rotations swing x_2 onto the position vector: first longitude θ about z_1, then latitude ϕ about y_2.

COMBINED DCM

$$[\hat{e}_2] = R_{y_2}(-\phi)\, R_{z_1}(\theta)\, [\hat{e}_1]$$

$$= \begin{bmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{bmatrix}$$

STEP 2: LATITUDE R_Y₂(−Φ)

$\phi = 2.9°$

$OX_2Y_2Z_2$ → $OX''Y''Z''$: Velocity-Referenced

$OX_2Y_2Z_2$ → $OX''Y''Z''$ — VELOCITY-REFERENCED

Dim purple = RST reference. Bright axes rotate in two steps: heading ψ about x_2, then flight-path γ about the new z'.

COMBINED DCM

$$[\hat{e}''] = R_{z'}(-\gamma)\, R_{x_2}(\psi)\, [\hat{e}_2]$$

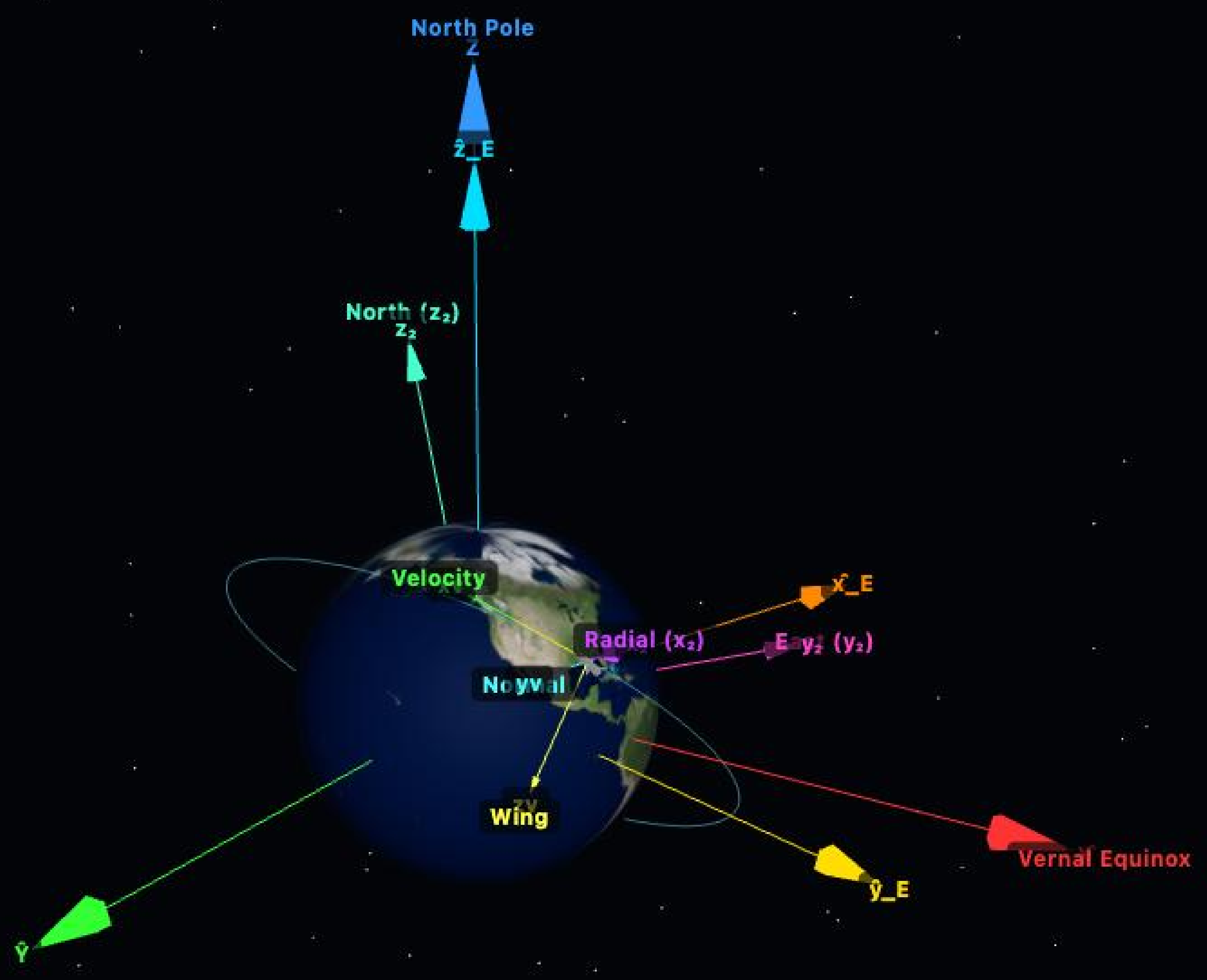
$$= \begin{bmatrix} \cos\gamma & -\sin\gamma\cos\psi & -\sin\gamma\sin\psi \\ \sin\gamma & \cos\gamma\cos\psi & \cos\gamma\sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix}$$

STEP 2: FLIGHT-PATH R_Z'(−Γ)

γ = 0.0°

$\hat{e}_{Y''}$ corresponds to $\hat{x}_v$ in this presentation.

Complete Transformation Chain



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COMPLETE TRANSFORMATION CHAIN

FULL DCM CHAIN

$$[\hat{e}''] = \underbrace{C_{OX_2\to OX''}}_{\text{attitude}}\,;\,\underbrace{C_{OX_1\to OX_2}}_{\text{position}}\,;\,\underbrace{C_{OXYZ\to OX_1}}_{\text{Earth spin}}\,;\,[\hat{e}]$$

EARTH SPIN — $C_{OXYZ\to OX_1}$

$$R_z(\theta_E) = \begin{bmatrix} \cos\theta_E & \sin\theta_E & 0 \\ -\sin\theta_E & \cos\theta_E & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

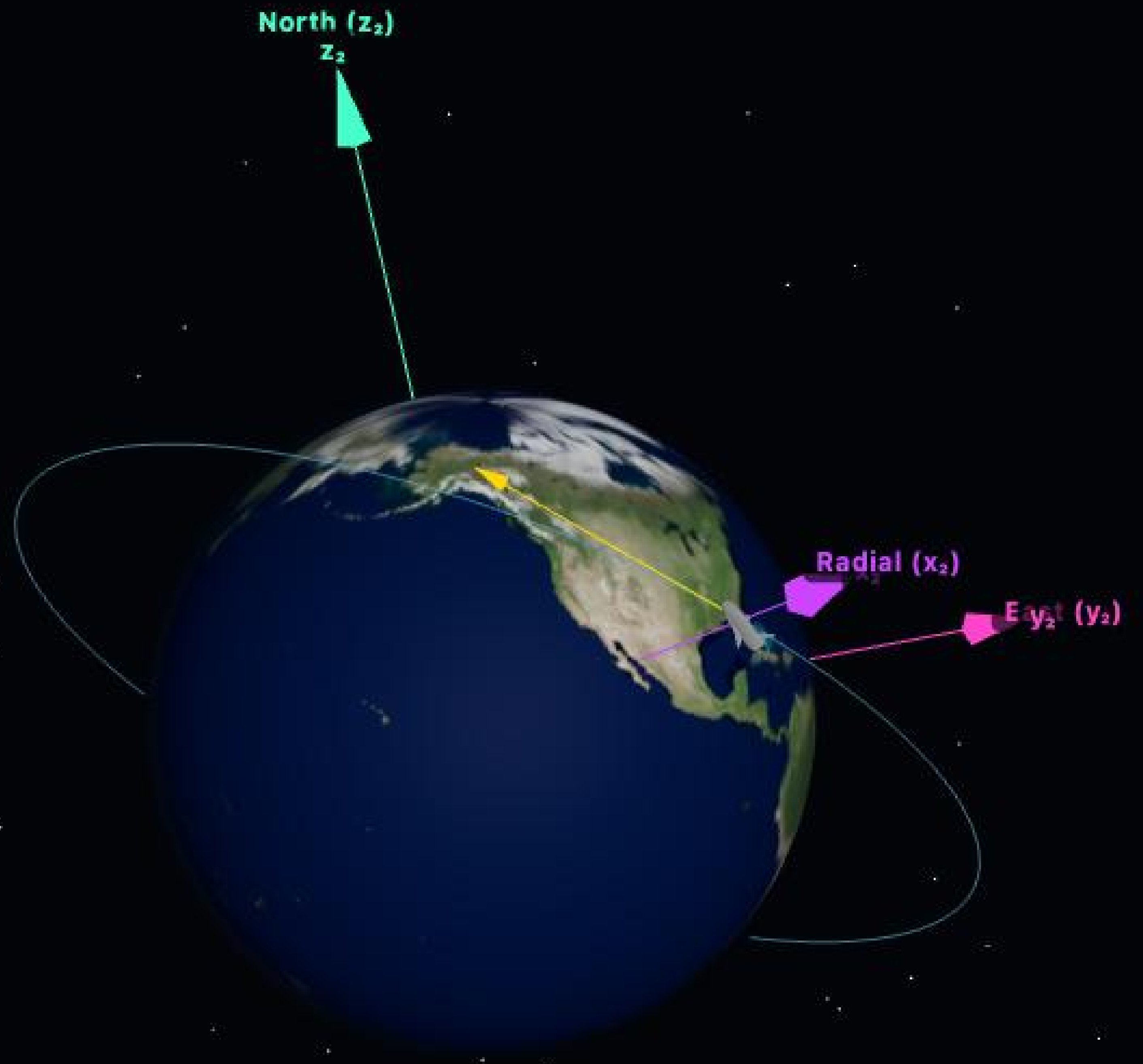
POSITION — $C_{OX_1\to OX_2}$

$$R_{y_2}(-\phi)\,R_{z_1}(\theta) = \begin{bmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{bmatrix}$$

ATTITUDE — $C_{OX_2\to OX''}$

$$R_{z'}(-\gamma)\,R_{x_2}(\psi) = \begin{bmatrix} \cos\gamma & -\sin\gamma\cos\psi & -\sin\gamma\sin\psi \\ \sin\gamma & \cos\gamma\cos\psi & \cos\gamma\sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix}$$

$C^{-1} = C^T$ — every DCM is orthonormal. To go OX″→OXYZ, transpose each matrix and multiply in reverse.



Kinematic Equations — $\dot{r}$, $\dot{\phi}$, $\dot{\lambda}$

The velocity vector projected onto the $OX_2Y_2Z_2$ axes links the six state variables: three from kinematics (position rates) and three from Newton's law (velocity rates).

ALTITUDE RATE — X_2 (RADIAL) COMPONENT

$$\dot{r} = v \sin \gamma$$

LATITUDE RATE — Y_2 COMPONENT

$$\dot{\phi} = \frac{v \cos \gamma \sin \psi}{r}$$

LONGITUDE RATE — Z_2 COMPONENT

$$\dot{\lambda} = \frac{v \cos \gamma \cos \psi}{r \cos \phi}$$

These three equations complete the six-state system once Newton's law supplies $\dot{v}$, $\dot{\gamma}$, $\dot{\psi}$. The $r\cos\phi$ denominator in $\dot{\lambda}$ accounts for meridian convergence — longitude circles shrink toward the poles.

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Newton's Second Law

With constant mass, Newton's Second Law in the inertial frame gives us the starting point for the entire EOM derivation.

INERTIAL EOM

$$m\ddot{\vec{r}}_I = \vec{F}_T + \vec{F}_{\text{aero}} + \vec{F}_g$$

FORCES ACTING ON THE VEHICLE

- Thrust — propulsive force
- Drag — aerodynamic retarding force
- Lift — aerodynamic perpendicular force
- Gravity — central body attraction

The superscript I on $\ddot{\vec{r}}$ emphasizes the derivative is taken with respect to the *inertial* frame.

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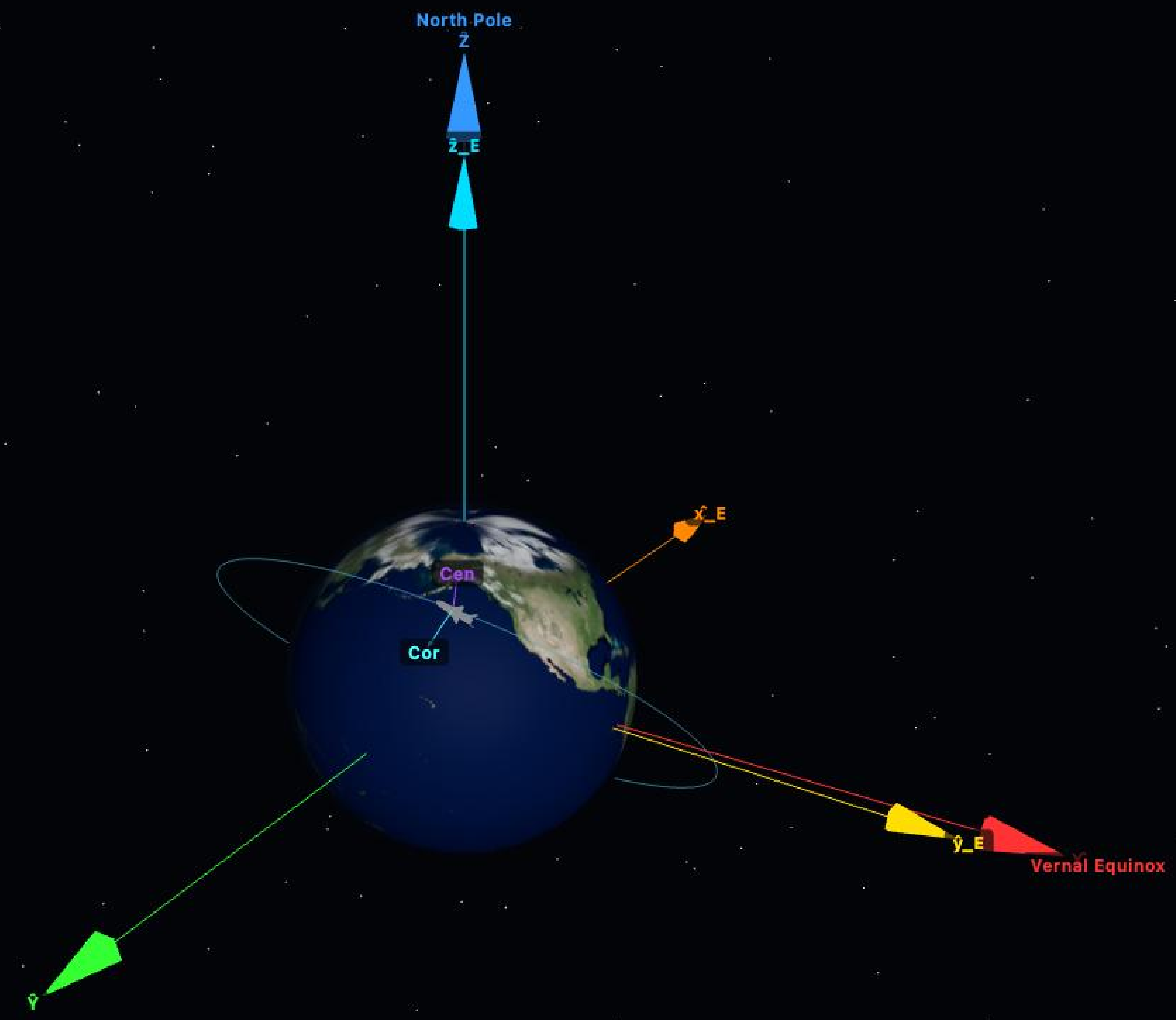


Rotating Frame Conversion

Reentry trajectory measurement is more convenient in the rotating, planet-fixed frame. Converting introduces two new acceleration terms.

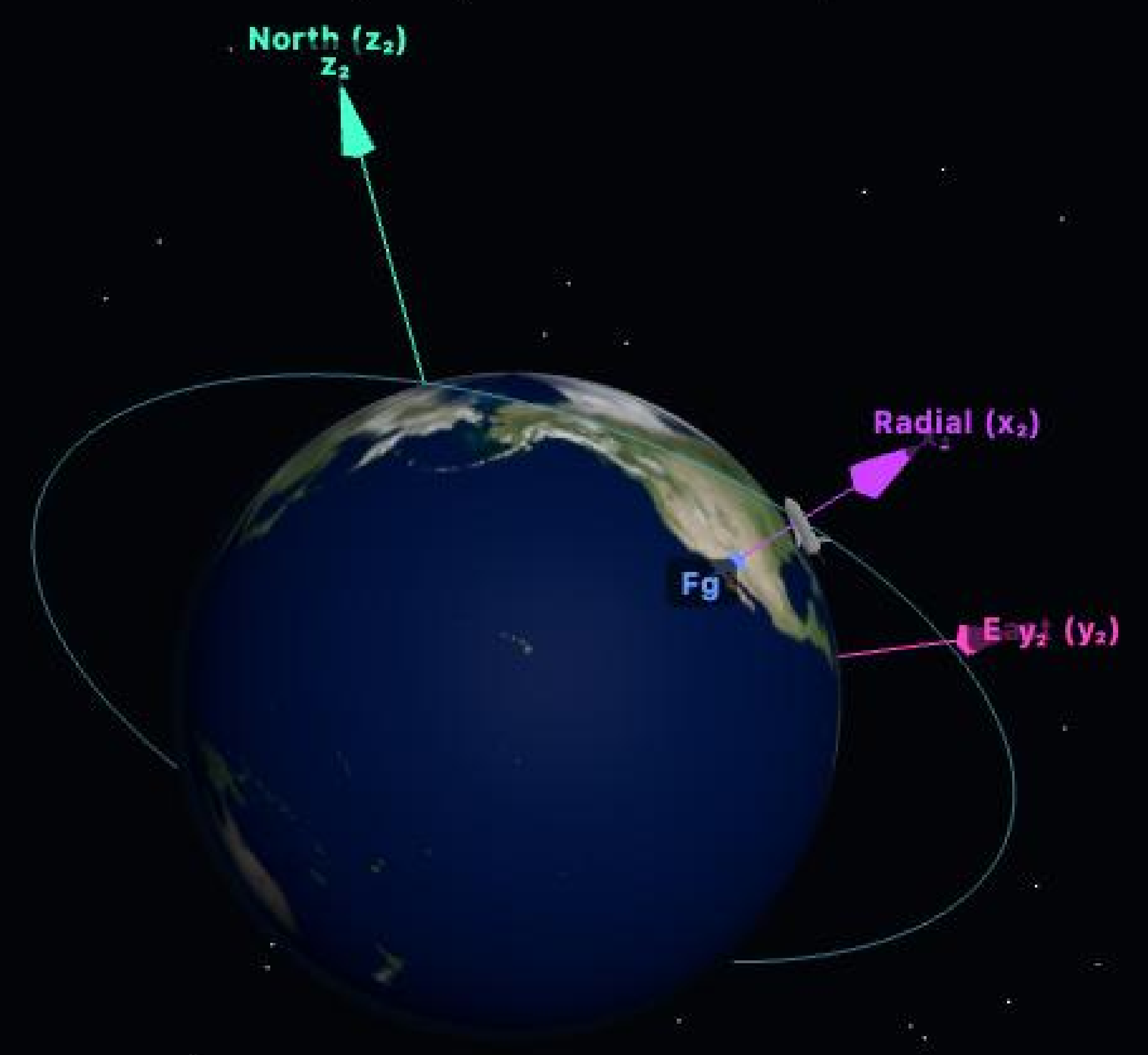
INERTIAL VS. ROTATING DERIVATIVE

$$\ddot{\vec{r}}_I = \underbrace{\ddot{\vec{r}}_{\text{rot}}}_{\text{rel. accel.}} + \underbrace{2\vec{\omega}_\oplus \times \dot{\vec{r}}_{\text{rot}}}_{\text{Coriolis}} + \underbrace{\vec{\omega}_\oplus \times (\vec{\omega}_\oplus \times \vec{r})}_{\text{centripetal}}$$

- Coriolis — depends on velocity relative to planet
- Centripetal — depends on distance from spin axis

Both terms update in real-time as the spacecraft moves — watch the arrows change direction along the orbit.

Gravitational Force



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Gravity acts radially inward along $-\hat{R}$ and follows the inverse-square law.

GRAVITATIONAL FORCE

$$\vec{F}_g = -\frac{\mu m}{r^2}\hat{R}$$

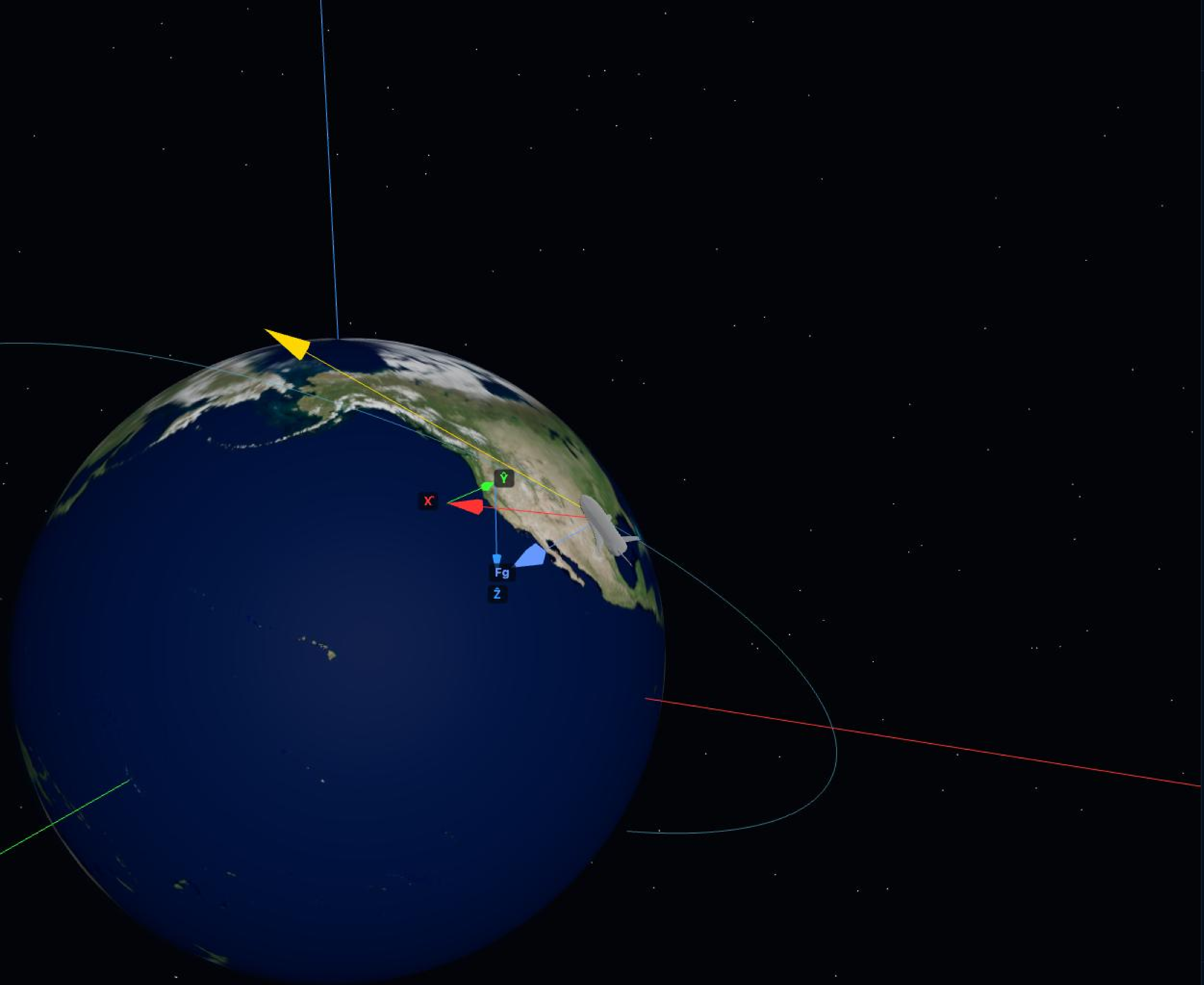
$$\mu = GM_{\oplus} = 3.986 \times 10^{14}\ \mathrm{m^3/s^2}$$

In the vehicle-pointing frame, the gravitational acceleration vector has only a radial component:

$$\vec{g} = -\frac{\mu}{r^2}\hat{R} = \begin{bmatrix} -g \\ 0 \\ 0 \end{bmatrix}_{\mathrm{RST}}$$

The gravity arrow grows longer as the spacecraft descends closer to Earth.

Gravity — ECI Decomposition



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GRAVITY — IN ECI

ROTATION CHAIN: RST → ECI

$$\vec{F}_g = C_{E\leftarrow R}\begin{bmatrix} -\mu m/r^2 \\ 0 \\ 0 \end{bmatrix}_{RST} = -\frac{\mu m}{r^2}\hat{R}$$

ECI COMPONENT BREAKDOWN

$$\vec{F}_g = -\frac{\mu m}{r^3}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{ECI}$$

$\hat{X}$ $\hat{Y}$ $\hat{Z}$

Drag Force

Drag acts directly *opposite* to the velocity vector — always in the $-\hat{x}_v$ direction of the VRF.

AERODYNAMIC DRAG

$$\vec{F}_D = -\frac{1}{2}\rho V^2 S C_D\,\hat{v}$$

- ρ — atmospheric density (decreases with altitude)
- V — speed relative to planet-fixed frame
- S — reference area
- C_D — drag coefficient

IN VEHICLE-POINTING FRAME

$$\vec{F}_D = \begin{bmatrix} -D\sin\gamma \\ -D\cos\gamma\cos\psi \\ -D\cos\gamma\sin\psi \end{bmatrix}_{RST}$$

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Drag — ECI Decomposition

DRAG — IN ECI

FULL DCM CHAIN: VRF → RST → ECI

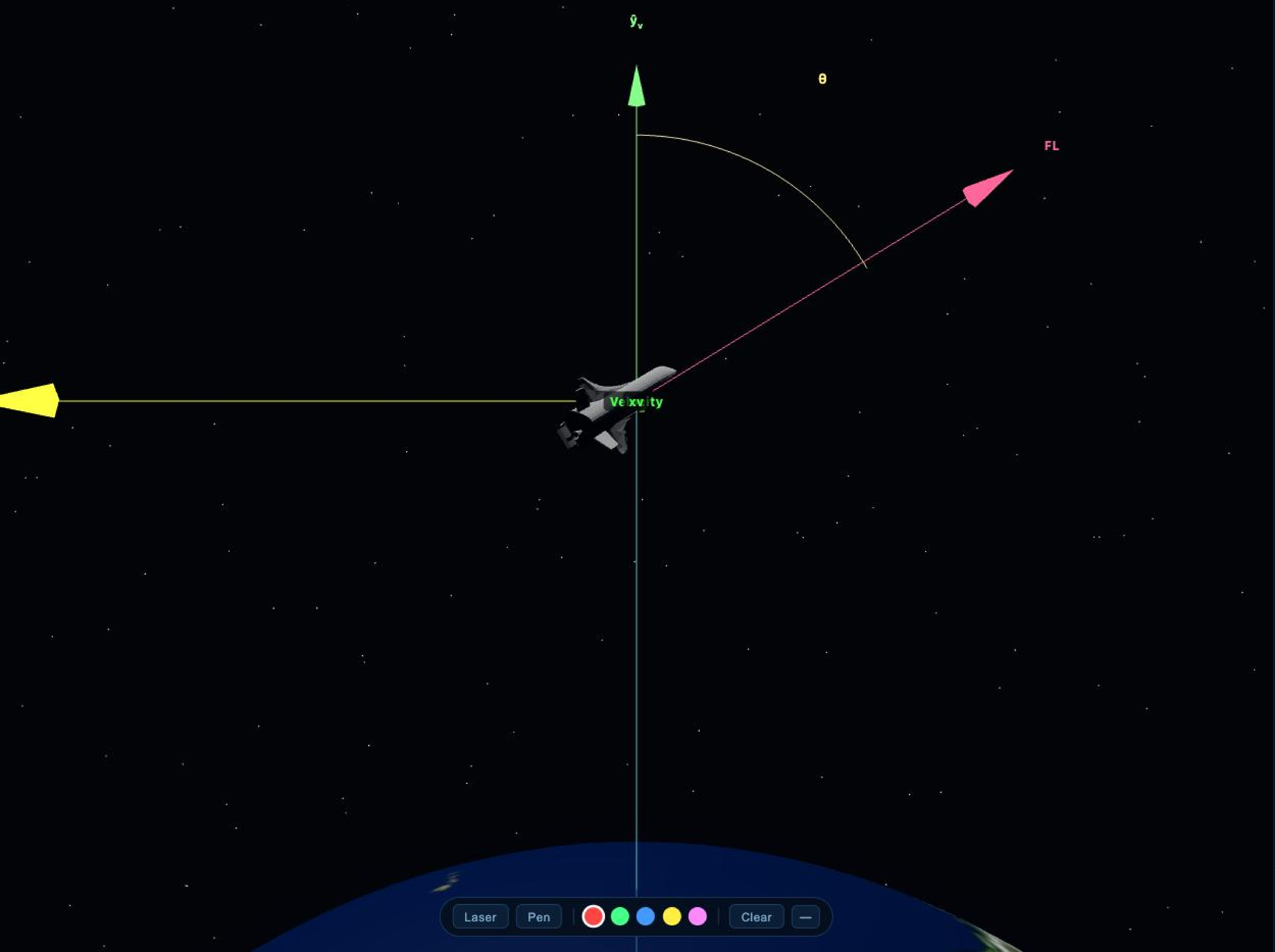
$$\vec{F}_D = C_{E\leftarrow R} \underbrace{\begin{bmatrix} \sin\gamma & \cos\gamma & 0 \\ \cos\gamma\cos\psi & -\sin\gamma\cos\psi & \sin\psi \\ \cos\gamma\sin\psi & -\sin\gamma\sin\psi & -\cos\psi \end{bmatrix}}_{C_{R\leftarrow V}} \begin{bmatrix} -D \\ 0 \\ 0 \end{bmatrix}_{VRF}$$

ECI COMPONENT BREAKDOWN

$$\vec{F}_D = \underbrace{C_{E\leftarrow R}(\phi,\theta)}_{\text{RST}\to\text{ECI}} \begin{bmatrix} -D\sin\gamma \\ -D\cos\gamma\cos\psi \\ -D\cos\gamma\sin\psi \end{bmatrix}_{RST} = \begin{bmatrix} F_{D,x} \\ F_{D,y} \\ F_{D,z} \end{bmatrix}_{ECI}$$

X̂ Ŷ Ẑ

Lift Force & Bank Angle θ



Lift is *perpendicular* to velocity. In the VRF (OX″Y″Z″), the **bank angle** θ rotates the lift vector about the velocity axis $\hat{x}_v$.

LIFT IN OX″Y″Z″ — BANK ROTATION ABOUT X_V

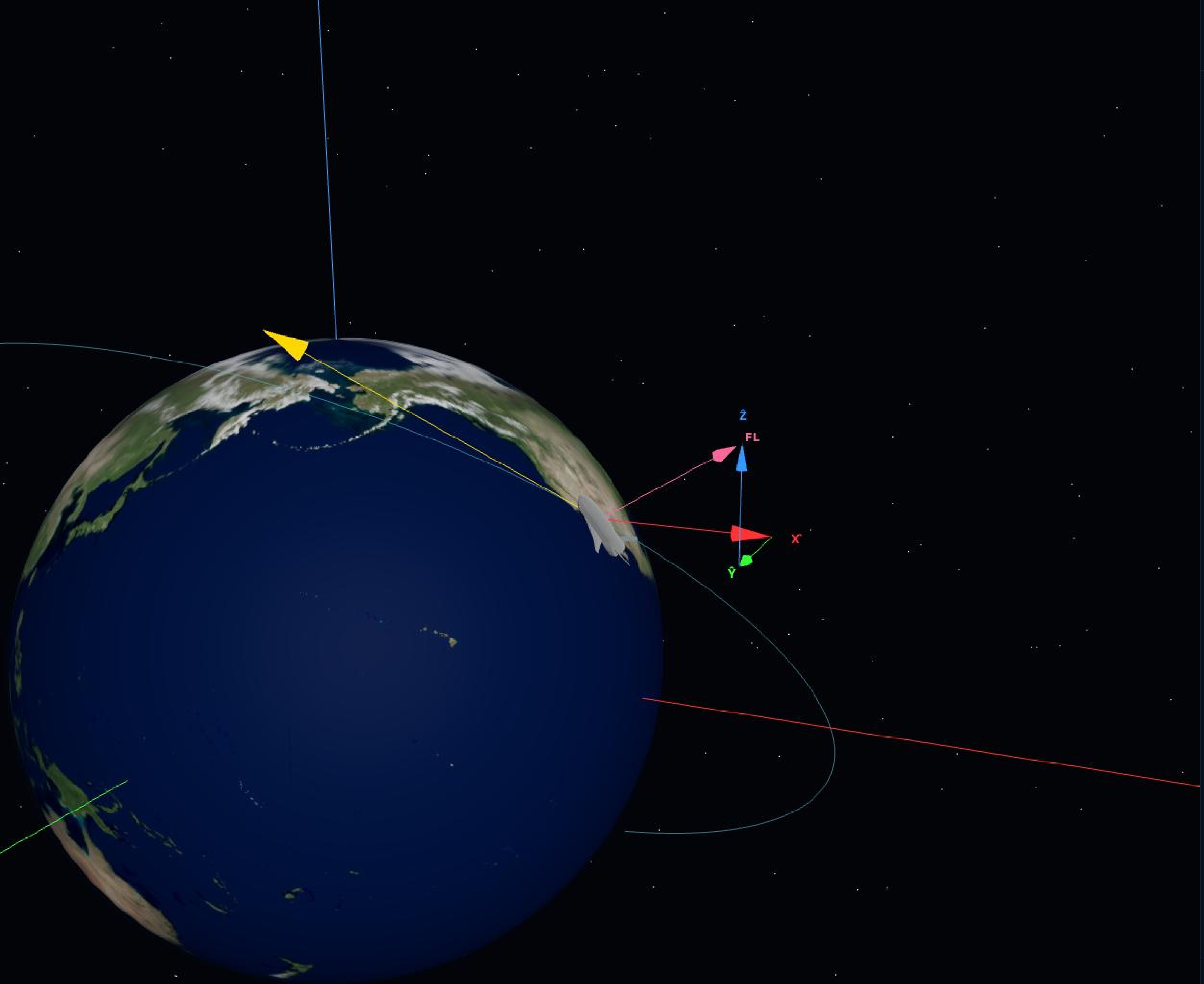
$$\vec{F}_L = L\begin{bmatrix}0\\ \cos\theta \\ \sin\theta\end{bmatrix}_{OX''Y''Z''} = L\left(\cos\theta\,\hat{y}_v + \sin\theta\,\hat{z}_v\right)$$

At $\theta = 0$, lift points along $\hat{y}_v$ (away from Earth, ⊥ to velocity). Banking sweeps it toward $\hat{z}_v$ (lateral), generating a cross-range acceleration component.

BANK ANGLE $\theta = 58.3°$

Varying θ is the primary guidance mechanism for lifting entry — it modulates both downrange and cross-range acceleration without changing lift magnitude.

Lift — ECI Decomposition



LIFT — IN ECI

FULL DCM CHAIN: VRF → RST → ECI

$$\vec{F}_L = C_{E\leftarrow R} \underbrace{\begin{bmatrix} \sin\gamma & \cos\gamma & 0 \\ \cos\gamma\cos\psi & -\sin\gamma\cos\psi & \sin\psi \\ \cos\gamma\sin\psi & -\sin\gamma\sin\psi & -\cos\psi \end{bmatrix}}_{C_{R\leftarrow V}} \begin{bmatrix} 0 \\ L\cos\theta \\ -L\sin\theta \end{bmatrix}_{VRF}$$

ECI COMPONENT BREAKDOWN

$$\vec{F}_L = \underbrace{C_{E\leftarrow R}(\phi,\theta)}_{\text{RST}\to\text{ECI}} \begin{bmatrix} -L\cos\theta\cos\gamma \\ L(\cos\theta\sin\gamma\cos\psi + \sin\theta\sin\psi) \\ L(\cos\theta\sin\gamma\sin\psi - \sin\theta\cos\psi) \end{bmatrix}_{RST} = \begin{bmatrix} F_{L,x} \\ F_{L,y} \\ F_{L,z} \end{bmatrix}_{ECI}$$

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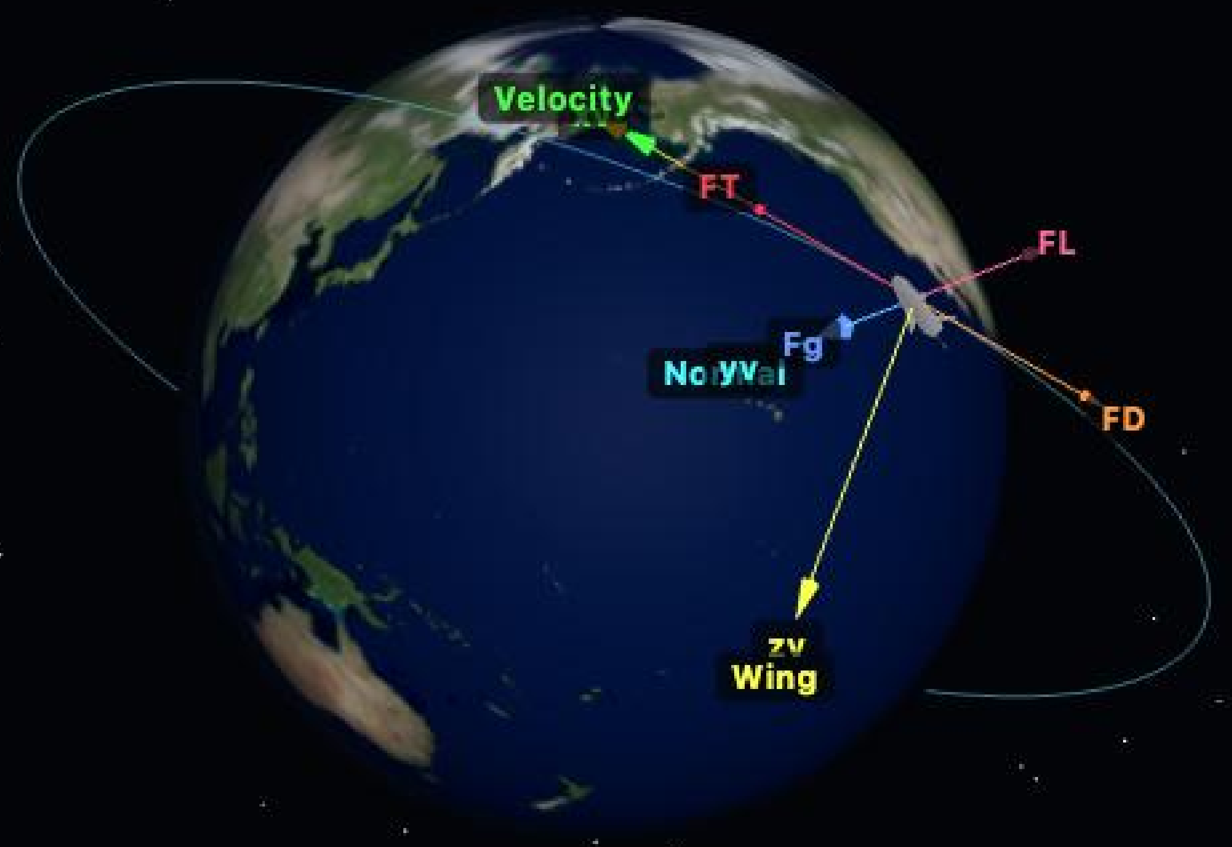
Thrust Force



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Thrust is expressed in the VRF using a pitch angle α_T and a bank angle β_T.

THRUST VECTOR (VRF)

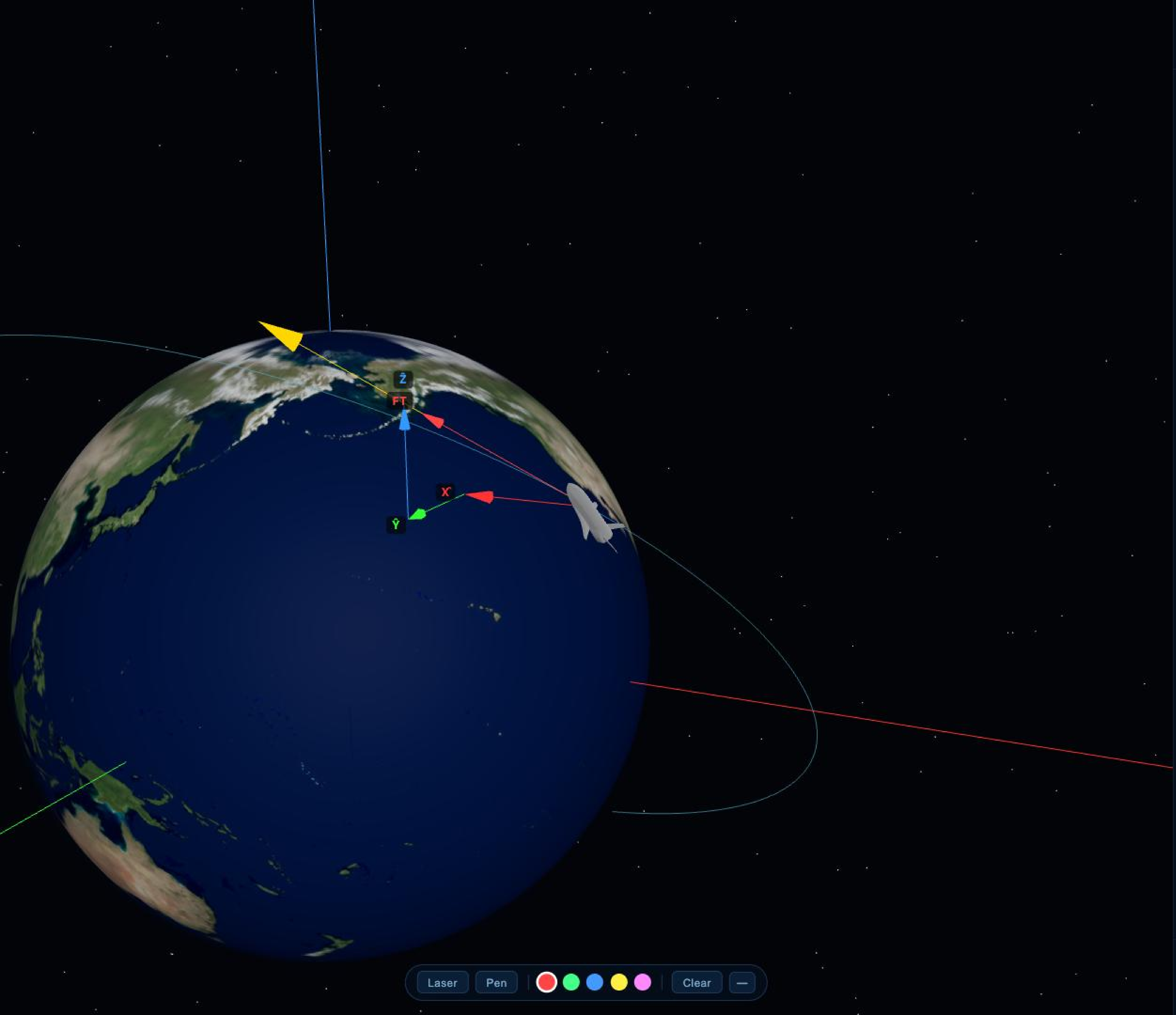
$$\vec{F}_T = T \begin{bmatrix} \cos\alpha_T \\ \sin\alpha_T \cos\beta_T \\ \sin\alpha_T \sin\beta_T \end{bmatrix}_{\text{VRF}}$$

Transforming to RST via $\mathbf{T}_{\text{VRF}\to\text{RST}}$ gives the thrust components used in the force equations.

All four forces are now simultaneously visible on the vehicle.

Hover to highlight in scene:

Gravity Drag Lift Thrust



Thrust — ECI Decomposition

THRUST — IN ECI

FULL DCM CHAIN: VRF → RST → ECI

$$\vec{F}_T = C_{E\leftarrow R}\underbrace{\begin{bmatrix} \sin\gamma & \cos\gamma & 0 \\ \cos\gamma\cos\psi & -\sin\gamma\cos\psi & \sin\psi \\ \cos\gamma\sin\psi & -\sin\gamma\sin\psi & -\cos\psi \end{bmatrix}}_{C_{R\leftarrow V}}\begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}_{VRF}$$

ECI COMPONENT BREAKDOWN

$$\vec{F}_T = \underbrace{C_{E\leftarrow R}(\phi,\theta)}_{\text{RST}\to\text{ECI}}\begin{bmatrix} T\sin\gamma \\ T\cos\gamma\cos\psi \\ T\cos\gamma\sin\psi \end{bmatrix}_{RST} = \begin{bmatrix} F_{T,x} \\ F_{T,y} \\ F_{T,z} \end{bmatrix}_{ECI}$$

X̂ Ŷ Ẑ

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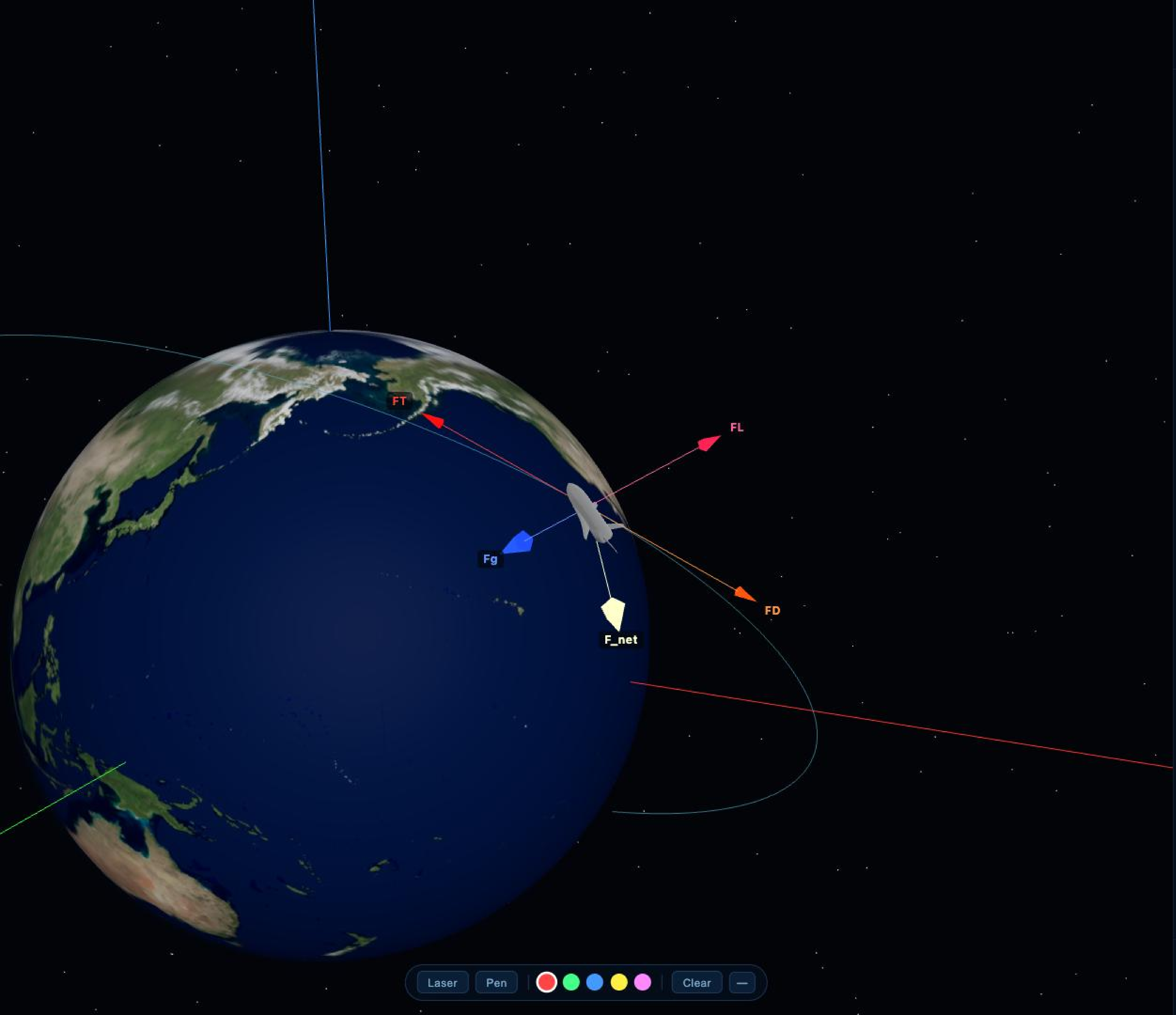

Newton's Law — All Forces in ECI

NEWTON'S LAW — ASSEMBLED IN ECI

All four forces share the same coordinate system and can be summed directly:

$$m\ddot{\vec{r}}_I = \vec{F}_g + \vec{F}_{aero} + \vec{F}_T$$

The cream arrow $\mathbf{F}_{\mathbf{net}}$ is their vector sum — ready to integrate for $r(t)$, $v(t)$, $\gamma(t)$, $\psi(t)$.

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F_net — ECI Components

F_{NET} — ECI COMPONENTS

All forces summed in ECI — each row is one axis component:

ECI COMPONENT BREAKDOWN

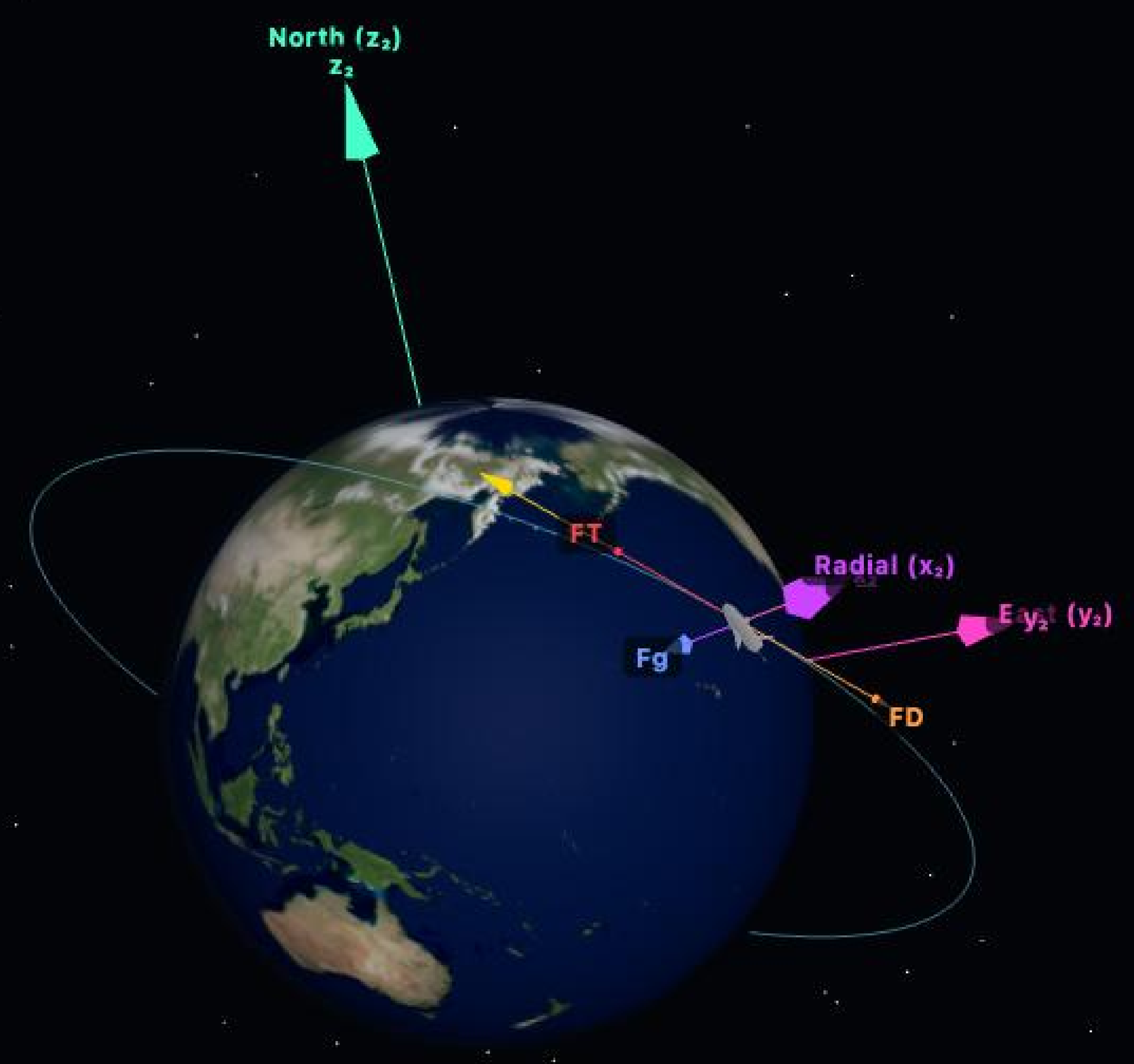
$$F_X = C_{E\leftarrow R}\left[-\frac{\mu m}{r^2} + (T-D)\sin\gamma - L\cos\theta\cos\gamma\right]$$

$$F_Y = C_{E\leftarrow R}\left[(T-D)\cos\gamma\cos\psi + L\cos\theta\sin\gamma\cos\psi + L\sin\theta\sin\psi\right]$$

$$F_Z = C_{E\leftarrow R}\left[(T-D)\cos\gamma\sin\psi + L\cos\theta\sin\gamma\sin\psi - L\sin\theta\cos\psi\right]$$

X̂ Ŷ Ẑ Then: $m\ddot{\vec{r}}_I = \vec{F}_{net}$

Force Assembly in $OX_2Y_2Z_2$



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All forces are expressed in $OX_2Y_2Z_2$ so their components can be summed axis-by-axis. Each RST component will produce one scalar ODE.

RADIAL X_2 — INWARD GRAVITY + DRAG/THRUST + LIFT

$$F_{x_2} = -\frac{\mu m}{r^2} + (T - D)\sin\gamma - L\cos\theta\cos\gamma$$

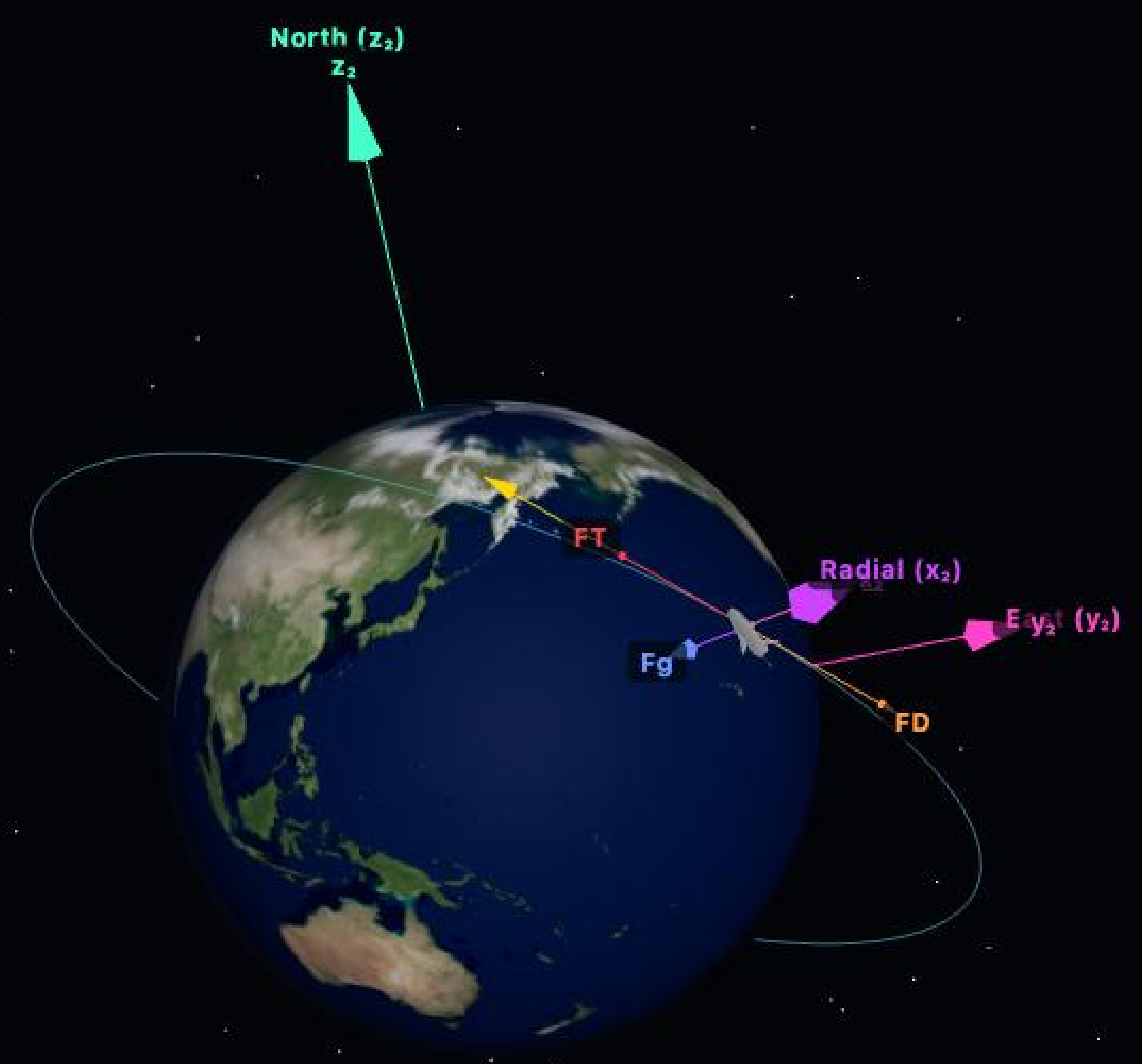
SOUTH Y_2 — DRAG/THRUST + LIFT MERIDIONAL

$$F_{y_2} = (T - D)\cos\gamma\cos\psi + L(\cos\theta\sin\gamma\cos\psi + \sin\theta\sin\psi)$$

EAST Z_2 — DRAG/THRUST + LIFT LATERAL

$$F_{z_2} = (T - D)\cos\gamma\sin\psi + L(\cos\theta\sin\gamma\sin\psi - \sin\theta\cos\psi)$$

Gravity Drag Lift Thrust



RST Projection → $\dot{v}$, $\dot{\gamma}$, $\dot{\psi}$

From slide 17: $m\ddot{\vec{r}}_I = m\ddot{\vec{r}}_{\text{rot}} + 2m\vec{\omega}_\oplus \times \dot{\vec{r}}_{\text{rot}} + m\vec{\omega}_\oplus \times (\vec{\omega}_\oplus \times \vec{r})$. Rearranging and projecting onto each RST axis:

X_2 (RADIAL) → $\dot{V}$

$$m\dot{v} = F_{x_2} + m\omega_\oplus^2 r\cos\phi(\sin\gamma\cos\phi - \cos\gamma\sin\phi\cos\psi)$$

Y_2 (SOUTH) → $\dot{\Gamma}$

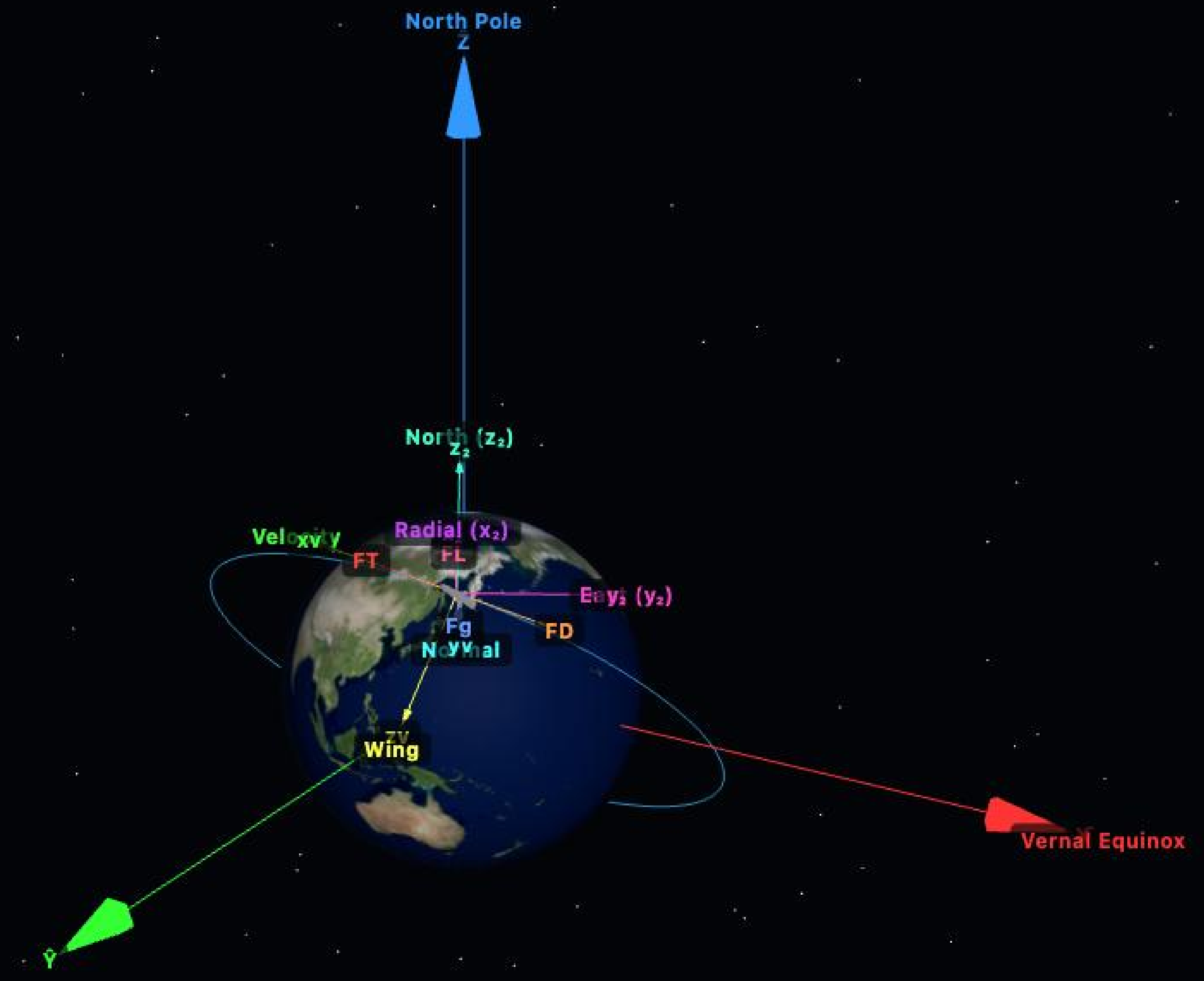
$$mv\dot{\gamma} = F_{y_2} - m\left(g - \frac{v^2}{r}\right)\cos\gamma + 2m\omega_\oplus v\cos\phi\cos\psi + m\omega_\oplus^2 r\cos\phi(\cos\gamma\cos\phi$$

Z_2 (EAST) → $\dot{\Psi}$

$$mv\cos\gamma\,\dot{\psi} = F_{z_2} + \frac{mv^2\cos^2\gamma\sin\psi\tan\phi}{r} - 2m\omega_\oplus v(\tan\gamma\cos\phi\cos\psi - \sin\phi) + \frac{m\omega}{}$$

Dividing through by m and mv, $mv\cos\gamma$ respectively gives $\dot{v}$, $\dot{\gamma}$, $\dot{\psi}$. Adding the three kinematic equations ($\dot{r}$, $\dot{\lambda}$, $\dot{\phi}$) completes the six-state system on the next slide.

Complete 3-DOF Equations of Motion



Six coupled scalar ODEs — complete **3-DOF point-mass** model.

KINEMATICS

$$\dot{r} = v\sin\gamma$$

$$\dot{\lambda} = \frac{v\cos\gamma\cos\psi}{r\cos\phi}$$

$$\dot{\phi} = \frac{v\cos\gamma\sin\psi}{r}$$

FORCES

$$\dot{v} = \frac{T\cos\alpha_T - D}{m} - g\sin\gamma + \omega_\oplus^2 r\cos\phi(\sin\gamma\cos\phi - \cos\gamma\sin\phi\cos\psi)$$

$$\dot{\gamma} = \frac{1}{v}\left[\frac{L\cos\theta}{m} - \left(g - \frac{v^2}{r}\right)\cos\gamma + 2\omega_\oplus v\cos\phi\cos\psi + \omega_\oplus^2 r\cos\phi(\cos\gamma\cos\phi + \sin\gamma\sin\phi\cos\psi)\right]$$

$$\dot{\psi} = \frac{1}{v\cos\gamma}\left[\frac{L\sin\theta}{m\cos\gamma} + \frac{v^2\cos^2\gamma\sin\psi\tan\phi}{r} - 2\omega_\oplus v(\tan\gamma\cos\phi\cos\psi - \sin\phi) + \frac{\omega_\oplus^2 r\sin\phi\cos\phi\sin\psi}{\cos\gamma}\right]$$

$$\mathbf{x} = (r,\, v,\, \gamma,\, \psi,\, \lambda,\, \phi)^T$$

Gravity Drag Lift Thrust

REFERENCE FRAMES

• ECI • RST • VRF

VECTORS

• Velocity • Gravity • Drag • Lift • Thrust

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Summary & Assumptions

The six-state system is complete. Here are the assumptions underlying the model:

- Entry vehicle treated as a **point mass**
- Planetary rotation is **constant** about the polar axis
- Planet-fixed and inertial frames **coincide at** $t_0 = 0$
- Vehicle mass is **constant** (no propellant depletion for EOM)
- Drag acts **opposite to velocity**
- Lift is **perpendicular to velocity**
- Gravity directed along $-\hat{R}$ (**no oblateness**)

COMPLETE STATE VECTOR

$$\mathbf{x} = \begin{bmatrix} r \\ v \\ \gamma \\ \psi \\ \lambda \\ \phi \end{bmatrix} \qquad \dot{\mathbf{x}} = f(\mathbf{x}, \vec{F})$$

Source: Hicks, K. D. *Introduction to Astrodynamic Reentry, 2nd ed.* 2014.

REFERENCE FRAMES

ECI | ECEF | RST | VRF

VECTORS

Velocity | Gravity | Drag | Lift | Thrust

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